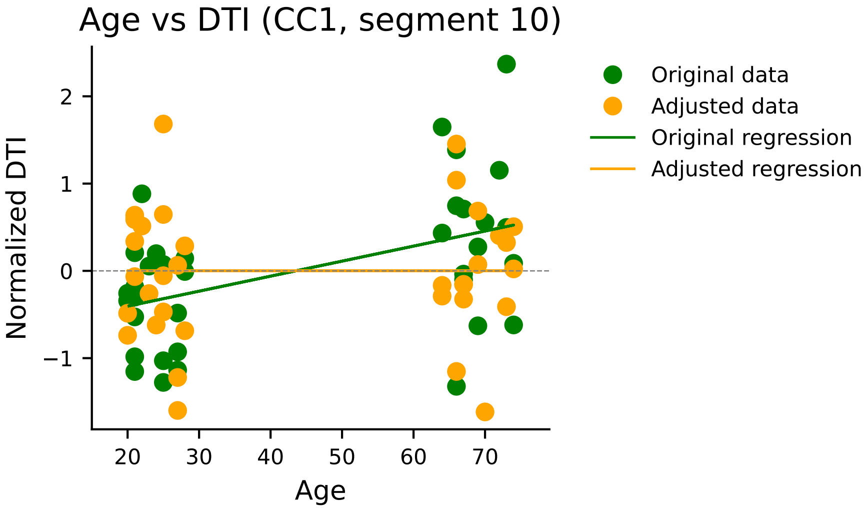
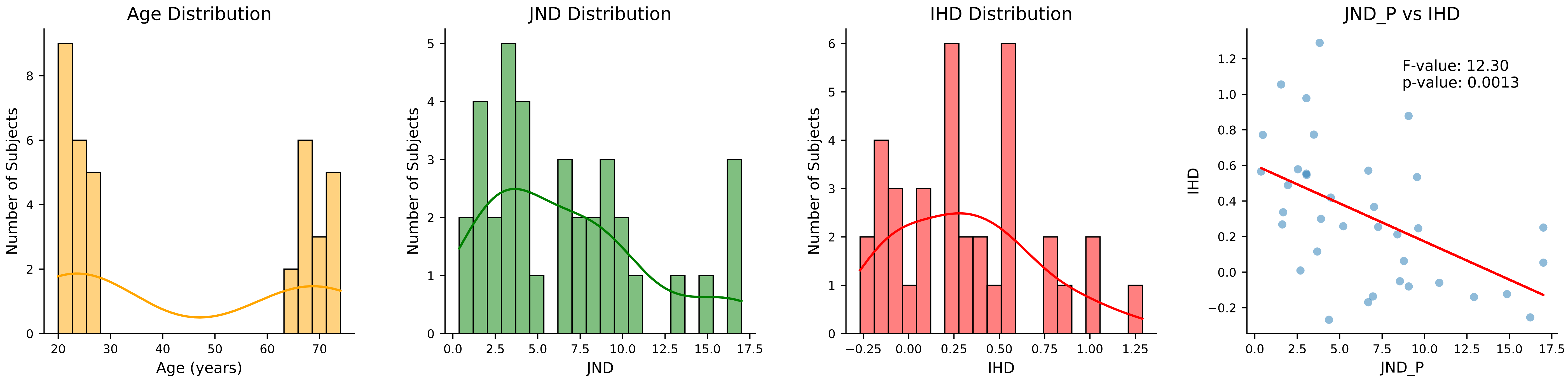
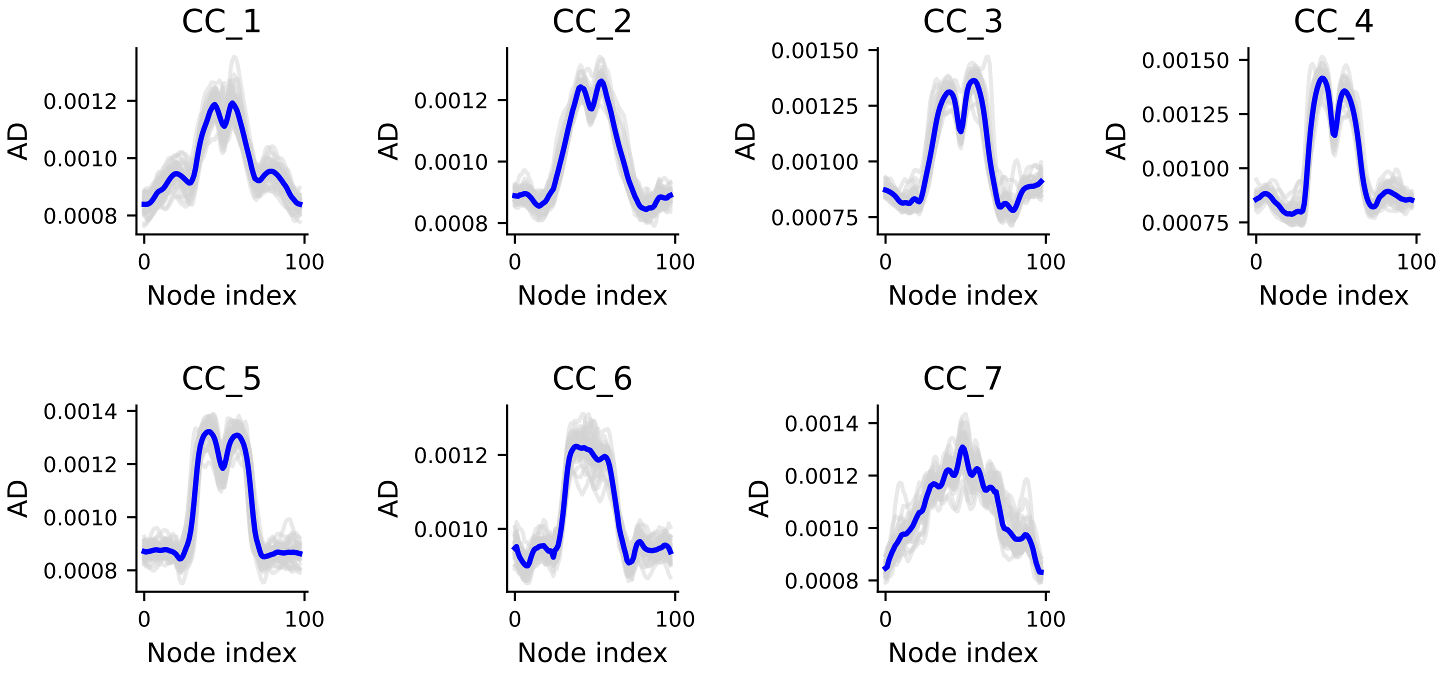
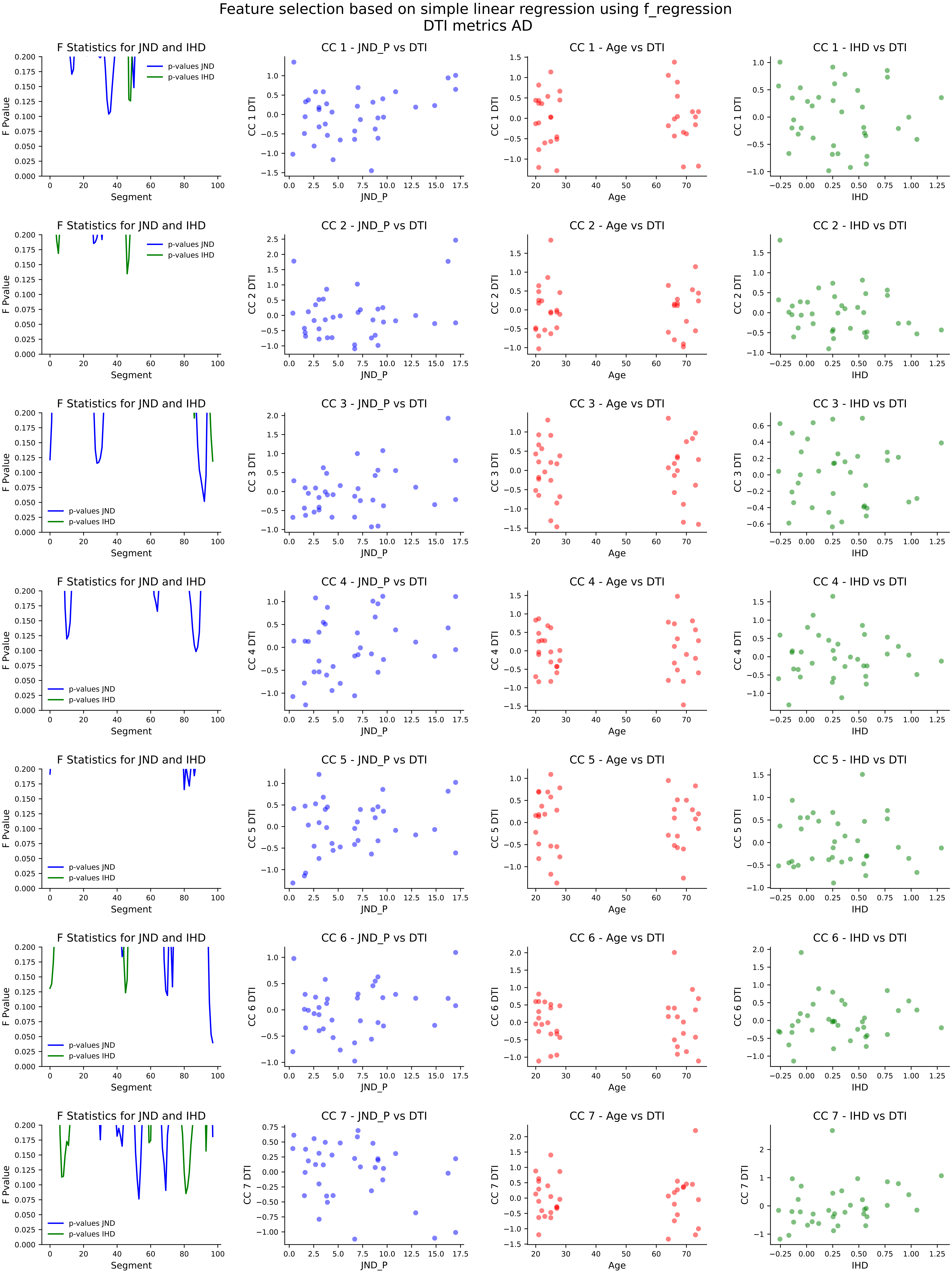


Original : p-value = 0.004, F-stat = 9.62

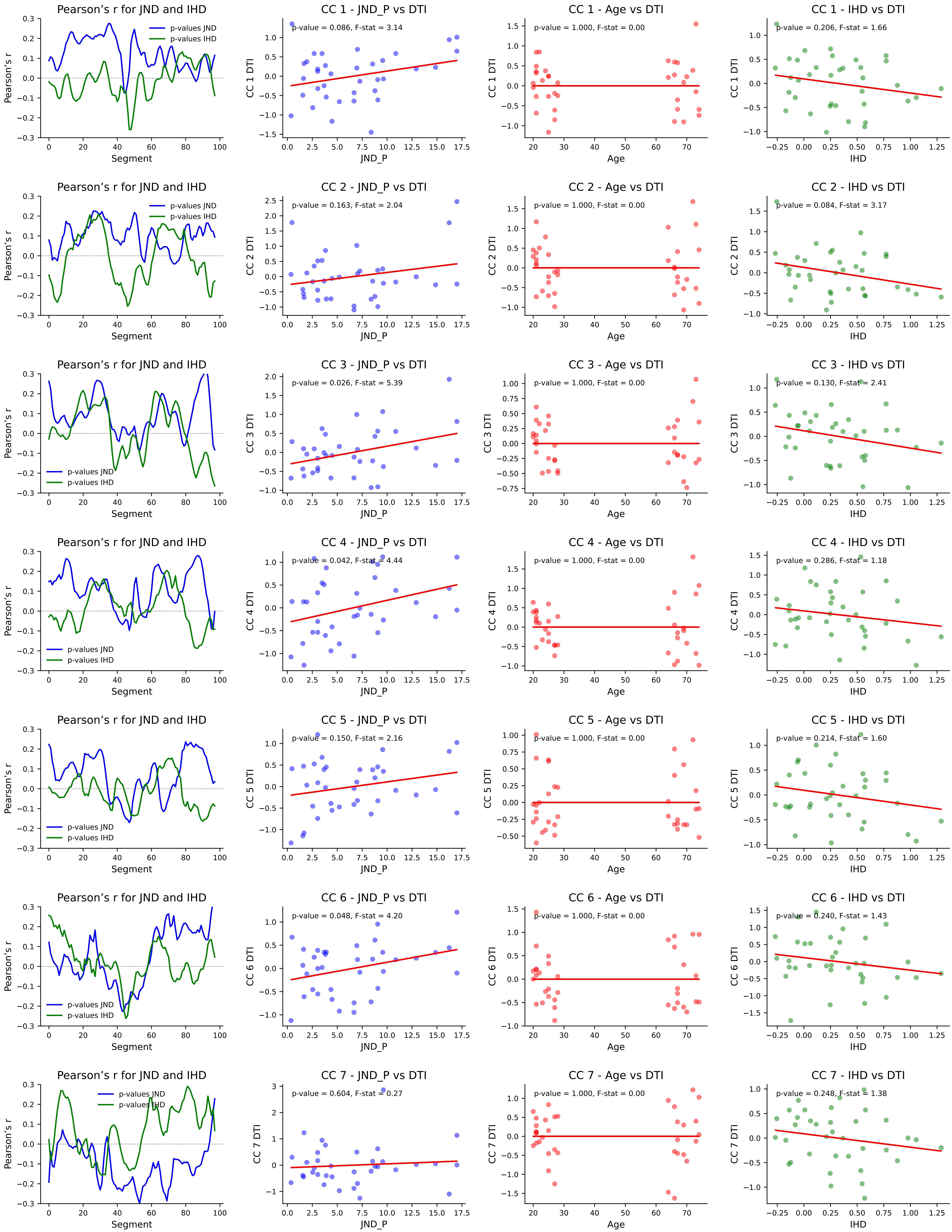
Adjusted : p-value = 1.000, F-stat = 0.00


Age Distribution
Number of Subjects
Age (years)
JND Distribution
JND
IHD Distribution
IHD
JND_P vs IHD
F-value: 12.30
p-value: 0.0013
IHD
JND_P

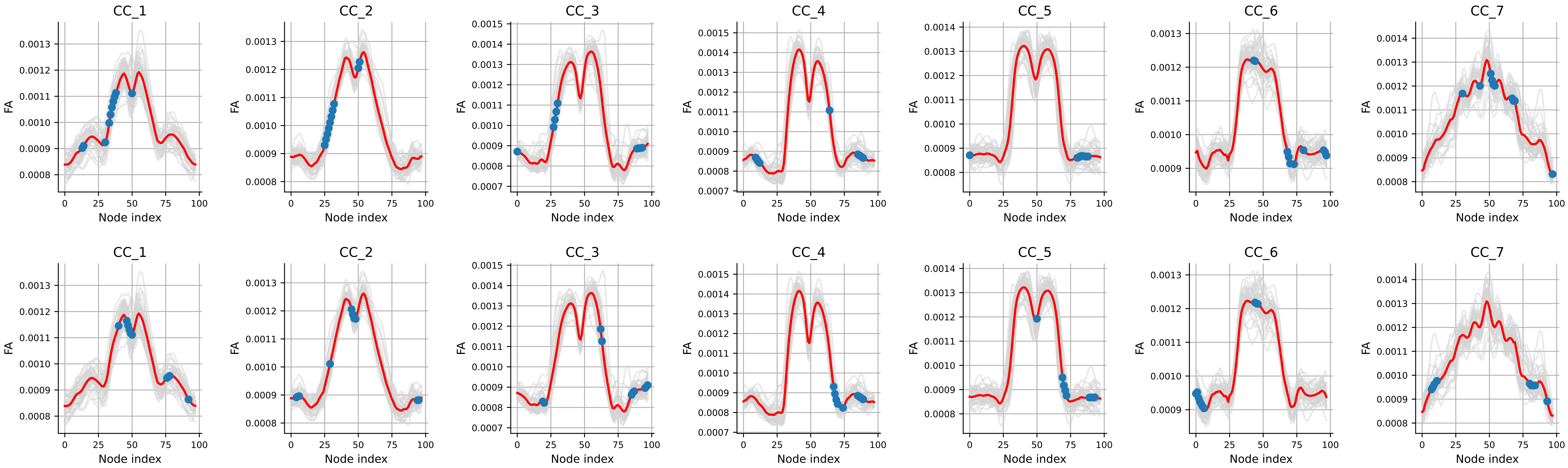

CC_1
CC_2
CC_3
CC_4
CC_5
CC_6
CC_7
AD
Node index
0
100
0.0008
0.0010
0.0012
0.0014
0.00075
0.00100
0.00125
0.00150

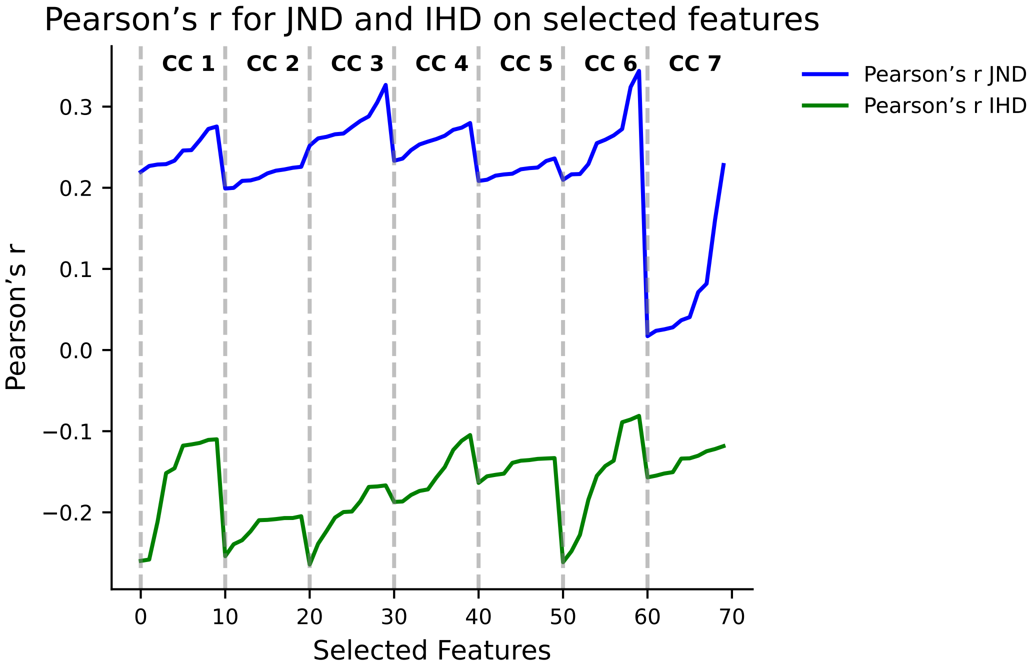

Feature selection based on simple linear regression using f_regression
DTI metrics AD
F Statistics for JND and IHD
CC 1 - JND_P vs DTI
CC 1 - Age vs DTI
CC 1 - IHD vs DTI
CC 2 - JND_P vs DTI
CC 2 - Age vs DTI
CC 2 - IHD vs DTI
CC 3 - JND_P vs DTI
CC 3 - Age vs DTI
CC 3 - IHD vs DTI
CC 4 - JND_P vs DTI
CC 4 - Age vs DTI
CC 4 - IHD vs DTI
CC 5 - JND_P vs DTI
CC 5 - Age vs DTI
CC 5 - IHD vs DTI
CC 6 - JND_P vs DTI
CC 6 - Age vs DTI
CC 6 - IHD vs DTI
CC 7 - JND_P vs DTI
CC 7 - Age vs DTI
CC 7 - IHD vs DTI
p-values JND
p-values IHD
F Pvalue
Segment
JND_P
Age
IHD
CC 1 DTI
CC 2 DTI
CC 3 DTI
CC 4 DTI
CC 5 DTI
CC 6 DTI
CC 7 DTI

Feature selection based on simple linear regression using r_regression
DTI metrics AD



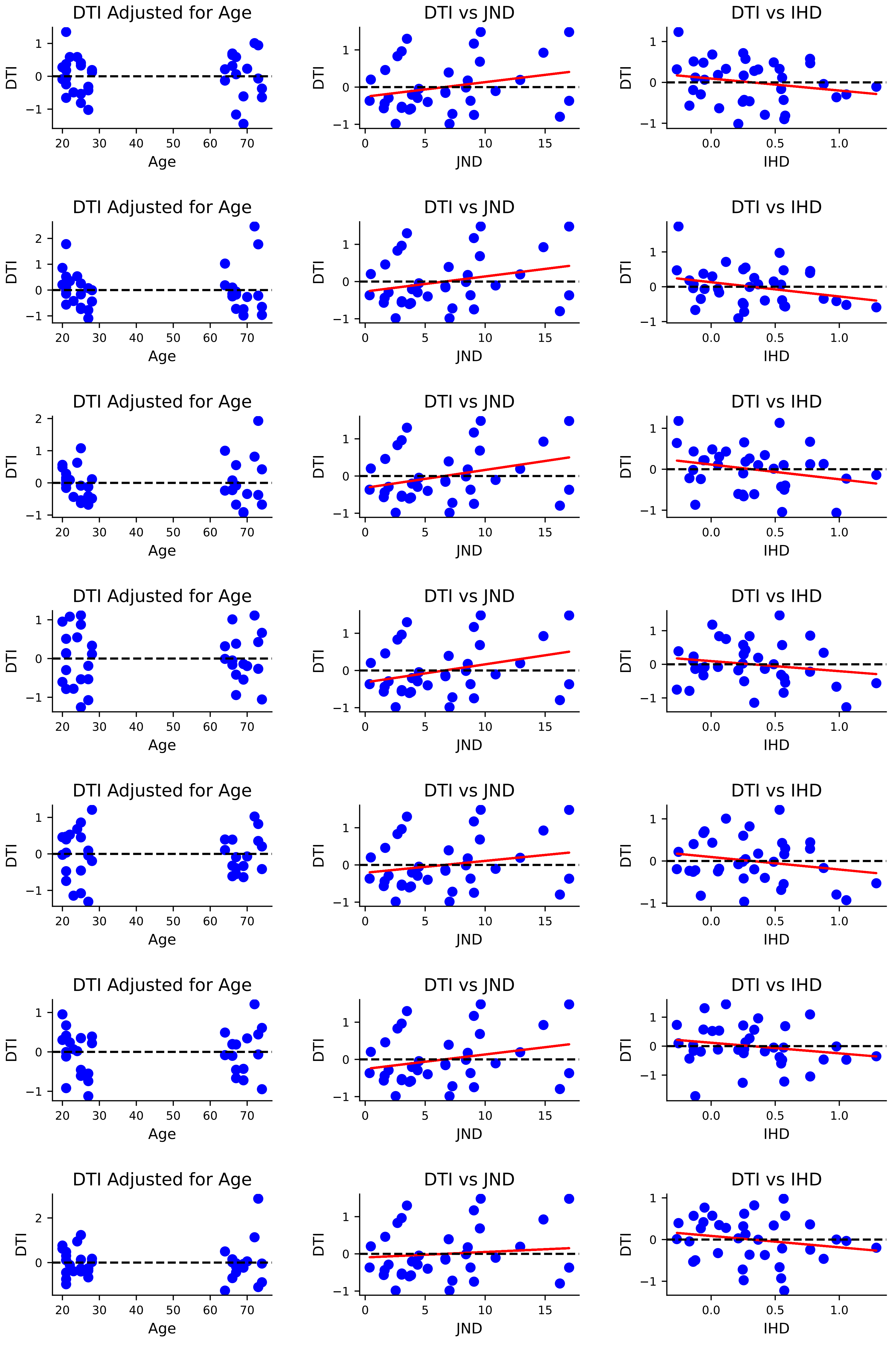
Best features for JND and IHD based





Pearson’s r for JND and IHD on selected features
CC 1
CC 2
CC 3
CC 4
CC 5
CC 6
CC 7
Pearson’s r JND
Pearson’s r IHD
Pearson’s r
0.3
0.2
0.1
0.0
−0.1
−0.2
0
10
20
30
40
50
60
70
Selected Features

Linear regression of DTI on JND and IHD after adjustment for age
Mean of 10 best features DTI metrics AD



JND AD

AD JND \ segment cc: 1 [35, 36, 34, 37, 50, 13, 14, 33, 38, 30]
p-value pour cc 1: 0.0236
p-value bonf. corrected pour cc 1: 0.165 □

AD JND \ segment cc: 2 [26, 27, 31, 28, 50, 29, 30, 51, 25, 32]
p-value pour cc 2: 0.0679
p-value bonf. corrected pour cc 2: 0.475 □

AD JND \ segment cc: 3 [92, 91, 90, 93, 89, 28, 29, 0, 30, 27]
p-value pour cc 3: 0.00308
p-value bonf. corrected pour cc 3: 0.022 □

AD JND \ segment cc: 4 [87, 88, 86, 10, 11, 89, 85, 12, 64, 9]
p-value pour cc 4: 0.00715
p-value bonf. corrected pour cc 4: 0.050 □

AD JND \ segment cc: 5 [80, 83, 82, 86, 0, 84, 81, 87, 85, 88]
p-value pour cc 5: 0.0598
p-value bonf. corrected pour cc 5: 0.419 □

AD JND \ segment cc: 6 [97, 96, 95, 70, 69, 73, 68, 43, 44, 80]
p-value pour cc 6: 0.00895
p-value bonf. corrected pour cc 6: 0.063 □

AD JND \ segment cc: 7 [53, 69, 52, 54, 68, 51, 67, 43, 30, 97]
p-value pour cc 7: 0.502
p-value bonf. corrected pour cc 7: 3.514 □

____________________IHD AD____________________

AD \ segment cc: 1 [48, 47, 49, 46, 50, 40, 16, 58, 57, 15]
p-value pour cc 1: 0.0967
p-value bonf. corrected pour cc 1: 0.677 □

AD \ segment cc: 2 [46, 47, 5, 4, 6, 48, 45, 94, 95, 3]
p-value pour cc 2: 0.0221
p-value bonf. corrected pour cc 2: 0.154 □

AD \ segment cc: 3 [97, 96, 86, 95, 87, 85, 39, 88, 55, 38]
p-value pour cc 3: 0.0456
p-value bonf. corrected pour cc 3: 0.319 □

AD \ segment cc: 4 [87, 86, 85, 88, 89, 90, 84, 91, 16, 17]
p-value pour cc 4: 0.162
p-value bonf. corrected pour cc 4: 1.131 □

AD \ segment cc: 5 [90, 89, 91, 92, 93, 50, 48, 88, 51, 49]
p-value pour cc 5: 0.102
p-value bonf. corrected pour cc 5: 0.717 □

AD \ segment cc: 6 [45, 46, 44, 47, 48, 43, 49, 85, 50, 84]
p-value pour cc 6: 0.123
p-value bonf. corrected pour cc 6: 0.859 □

AD \ segment cc: 7 [31, 33, 32, 30, 29, 47, 48, 50, 49, 26]
p-value pour cc 7: 0.129
p-value bonf. corrected pour cc 7: 0.906 □